

Comparative Impact of AI-Based Precision Scheduling & Solar Irrigation on Agricultural Greenhouse Gas Emissions

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Abstract:

Agriculture accounts substantially to global greenhouse gas (GHG) emissions due to fossil fuel use, inefficient irrigation, and input mismanagement. Technologies such as artificial intelligence and renewable energy systems highlight powerful solutions to improve efficiency and sustainability. This study investigates which specific technological factors have the greatest impact on reducing greenhouse gas emissions in agricultural systems.

Simulated data was generated to model the performance of four farm scenarios: a control using conventional diesel-powered irrigation, a group only using AI based irrigation scheduling, a group only using solar powered irrigation, and a group combining both AI and solar technologies. All were evaluated over a 120 day growing season under uniform crop, soil, and climate conditions. Metrics included net GHG emissions, water-use efficiency, resource consumption, and crop yield. Emissions were calculated using the KGML-ag-Carbon model⁶, which estimates net ecosystem carbon exchange and changes in soil organic carbon.

The integrated system achieved the greatest GHG reduction, highlighting the benefits of combining AI and solar power. These findings provide insight into optimizing technological interventions for climate smart agriculture.

Introduction:

Greenhouse gas emissions from agriculture continue to rise due to diesel powered equipment, poor irrigation practices, and soil degradation. Researchers are turning to technologies that can reduce emissions while maintaining productivity, specifically AI and renewable energy systems. In greenhouse studies, AI has reduced heating energy use, and in some cases, water consumption¹⁻⁷. Additionally, powered irrigation systems, especially when paired with IoT devices, decreased diesel dependency and minimize emissions in field agriculture^{2,4}. Tools like KGML-ag-Carbon⁶ provide advanced modeling frameworks to estimate emissions from agroecosystems by simulating carbon flows in soil and plant systems.

AI also improves renewable energy system performance by improving solar energy efficiency through predictive maintenance, load balancing, and smart energy distribution³. While studies show that AI and renewable energy technologies can support emission reduction efforts individually^{1,2,5}, few examine how specific integrated technological factors can affect outcomes. This study addresses that gap by investigating how various applications differ in their effectiveness. The objective of this study is to determine which technological factor has the greatest impact on reducing GHGs in agricultural systems. To hypothesize, if solar-powered irrigation systems are combined with AI-driven precision scheduling of agricultural operations, then farms will show a greater reduction in greenhouse gas emissions compared to those using either AI or solar power independently.

Methods:

Simulated datasets were created for four groups of mid-sized farms (~100 hectares each), all with standardized soil type, crop variety (corn), and semi-arid climate. The control group will represent farms using diesel-powered irrigation systems with no AI optimization. Group A will include farms using AI based crop and irrigation management only, featuring scheduling algorithms that adjust water and input use based on weather forecasts and soil moisture data^{1,3}. Group B only uses solar-powered irrigation systems, supplying 100% of energy needs through photovoltaic panels². Group C combines AI precision scheduling and solar powered irrigation. Each group will be simulated over one 120 day growing season. Throughout the simulation, four key metrics will be tracked: net GHG emissions (kg CO₂-eq/ha), total fuel or electricity consumption (kWh/ha), water use efficiency (kg yield/m³ water), and total crop yield (tons/ha). Net ecosystem carbon exchange and soil organic carbon changes will be estimated using KGML-ag-Carbon model^{6,7} to provide daily carbon flow estimates. Each group will undergo 50 simulation trials to account for variability and statistical significance will be assessed using ANOVA and post-hoc tests ($\alpha = 0.05$). By comparing average emissions and resource usage across the four groups, the study will evaluate which specific technological intervention produces the most reductions in GHG emissions.

The simulated data represent estimates for key metrics such as greenhouse gas emissions, fuel or electricity consumption, water usage, and crop yield over a single growing season. These values

were generated using baseline figures from agricultural studies ^{1,2,3,6}, including average diesel usage for irrigation, efficiency improvements from AI-based scheduling, and energy outputs from solar-powered systems. For each group, performance was projected based on realistic assumptions and variability was introduced to reflect natural fluctuations in farming conditions.

Conclusion:

This study explored which technological interventions most effectively reduce GHG emissions in agriculture. Results from simulated farm trials showed that the combined use of AI precision scheduling and solar powered irrigation led to the greatest emission reductions, out performing each technology used alone. While simulations offer strong insights, limitations of this study include implementation costs and real world variability. Future studies should validate these findings in field conditions and explore region specific adaptations to ensure scalable and climate resilient agricultural solutions.

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Definition Sheet:

GHG (Greenhouse Gas) = Gases that trap heat in the atmosphere, such as carbon dioxide(CO₂), methane(CH₄), and nitrous oxide(N₂O). These gases contribute to global warming and climate change.

AI (Artificial intelligence) = Computer systems or algorithms designed to perform tasks that typically require human intelligence, such as decision making, pattern recognition, and data analysis.

Precision Agriculture = A farming management concept that uses data driven technologies to optimize field level management regarding crop farming

Renewable Energy = Energy from natural sources that are continuously replenished like solar or wind power

Solar Powered Irrigation = An irrigation system that is powered entirely by solar panels, which reduce reliance on fossil fuels and emissions associated with conventional diesel pumps

Irrigation Scheduling = The process of determining the optimal timing and quantity of water application to crops, in cases enhanced by AI for improved water usage efficiency

Diesel Powered = Vehicles, machinery, or equipment that run on diesel engines

IoT (Internet of Things) = A network of interconnected devices that collect and exchange data. In agriculture, IoT devices monitor soil moisture, weather, and crop health to improve decision making

KGML-ag-Carbon = A knowledge based machine learning model used to estimate greenhouse gas emissions in agroecosystems through simulating net ecosystem carbon exchange and soil organic carbon changes

Net Ecosystem Carbon Exchange = The net exchange of CO₂ between the ecosystem and atmosphere, reducing the atmospheric carbon dioxide concentrations

ANOVA (Analysis of Variance) = A test used to determine differences between research results from three or more unrelated samples or groups